

Contents

WEARABLES

18 No longer geeky, wearables are style statements

INCLUDES 28-PAGE SPECIAL SUPPLEMENT:



EDITOR : Ramesh Chopra

EDITORIAL
CORRESPONDENCE Phone: 011-26810601, E-mail: editor@efy.in
(Technical queries: eflyab@efy.in)

SUBSCRIPTIONS &
MISSING ISSUES Phone: 011-26810601 or 02 or 03
E-mail: support@efy.in

BACK ISSUES,
BOOKS, CDs,
PCBs etc. Phone: 011-26371661, 26371662
E-mail: info@kitspares.com

NEWS/STAND
DISTRIBUTION Ph: 011-40596600
E-mail: eflydc@efy.in

ADVERTISEMENTS
NEW DELHI (HEAD OFFICE) Ph: 011-26810601 or 02 or 03
E-mail: eflynm@efy.in

MUMBAI Ph: 022-24950047, 24928520
E-mail: eflymum@efy.in

BENGALURU Ph: 080-2526394, 25260023
E-mail: eflybln@efy.in

PUNE Ph: 08800295610, 09870682995
E-mail: eflypune@efy.in

GUJARAT Ph: 079-61344948
E-mail: eflyahd@efy.in

JAPAN Ph: 81-3-3541-4166
E-mail: japan@efy.in

SINGAPORE Ph: 65-6836 2272
E-mail: singapore@efy.in

TAIWAN Ph: 886-2-87726780 ext. 10
E-mail: taiwan@efy.in

UNITED STATES Ph: +1 860 536 6677
E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at International Press o-Pad Ltd, C-4 to C-11, Haryana Complex, Phase-II Extension, MODA-201 305, Gurgaon (Haryana), India. Published on the first day of each month and published from D-017, DDA Industrial Area, Phase-2, New Delhi 110028. Copyright 2018. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss or damage by publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFLY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in New Delhi court only.

SUBSCRIPTION RATES

Period	Newstand Price	You Pay	Digital	Overseas
Year	(₹)	(₹)	(₹)	Digital Print
Three	3600	2520	910	—
Two	2400	1800	648	US\$ 15
One	1200	960	345	US\$ 120

Please send payments only in favour of EFLY Enterprises Pvt Ltd



To know more details turn to Page No. 06

www.covestro.in

SMART WORLD

- 76 Corporate security: Digital is the way to go
- 78 Driverless cars: Automotive sector's next big business bet
- 81 INTERVIEW: QualityKiosk Technologies: "Test automation reduces quality assurance lifecycle by at least 50 per cent"

RESEARCH & DEVELOPMENT

- 28 AGRITECH: Nanosensors for agricultural pest management
- 32 MATERIALS: Electronics of exotic materials
- 36 DESIGN: High-frequency issues in electronics
- 42 SOLAR: Next-generation solar power technology (Part 2 of 2)
- 46 DESIGN: Designing electro-medical equipment
- 50 INNOVATION: Cloud-based IoT monitoring of pillbox usage
- 67 INTERVIEW: Nicolas Touchard, vice president, marketing, DxOMark: "Moving towards smartphone cameras that capture the same quality photographs as DSLR cameras"
- 68 INTERVIEW: Sandeep Agarwal, senior vice president and head of Global Center of Excellence for IoT (COE-IoT), Happiest Minds Technologies: "The IoT is not something which an SI partner will deploy and stop"

GUIDE

- 53 BUYER'S GUIDE: Microwave and PIR sensors: A small investment for big savings
- 61 Test & Measurement: Selecting the right T&M equipment for medical electronics

EVERYTHING YOU WANTED TO KNOW ABOUT...

- 55 NEPCON: Why I felt hot in Japan even in January
- 58 SENSORS: Embedded sensors for critical applications
- 65 WIRELESS: Wireless technology for medical applications

Next Issue

- Hyper convergence
- Software-centric
- Energy storage
- Sensors for industrial applications



To know more details turn to Page No.127

www.metroq.in

GREEN TECHNOLOGY

- 70 The power of water unchained, unabated

MAKE IN INDIA

- 86 STARTUP ZONE: Ink from DG sets; BlackBox to analyse vehicle accidents; IIoT solutions
- 90 MARKET SURVEY: Will biosensors sensitise the Indian market?
- 95 PROJECT REPORT: Business setup for LED lighting assembly

INDUSTRY

- 98 INDIA ELECTRONICS WEEK: Third edition of IEW shows the way ahead for IoT and startups

DO-IT-YOURSELF



- 102 Virtual telepresence robot using Raspberry Pi
- 108 Bi-directional motor driver using ULN2803
- 110 Automatic light and overcharging control with voltage-level indicator
- 113 Little nightlight
- 114 Automatic water refiller for air-coolers
- 116 Software-defined radio with Android smartphones
- 121 Audio compression using wavelets in MATLAB

NEWS

- 12 Tech News
- 83 Industry News
- 92 New Products

EFY Plus DVD

EFY + DVD: Useful tools and easy project management suites 82
Quite Universal Circuit Simulator: A Tool that Justifies its Name I
ProjectLibre: An Excellent Tool for Project Management IV
Simulink 0.15: Have Fun While Getting your Work Done VII

Regulars

- 08 Feedback: Your suggestions
- 10 Q&A: Things you wanted to know